

PerC-SimAM-BLSTM: Position Perception Circular Convolution With Simple Attention Mechanism Based on BLSTM for Bundle Branch Block Detection

Jibin Wang^{ID} and Bo Shi^{ID}

Abstract—Left bundle branch block (LBBB) and right bundle branch block (RBBB) are two typical types of arrhythmias. Due to their similar electrocardiogram (ECG) features, the conventional visual inspection ECG strategy has always been challenging. Motivated by MetaFormer, we investigate a novel MetaFormer-like model that is composed of both position perception circular convolution (PerC) as a new Token Mixer and simple attention mechanism with bidirectional long short-term memory (SimAM-BLSTM) as a new Channel MLP called PerC-SimAM-BLSTM for automatic LBBB and RBBB detection. This framework not only inherits the global feature extraction ability of Transformer but also enhances global-local feature acquisition capability of PerC. Experiment results demonstrate better improvements than the existing strategies and also strike a better balance between performance and complexity from an empirical experiment perspective. Mathematical convergence proof ensures stable convergence of model training to the theoretical optimal solution from a theoretical perspective. Our PerC-SimAM-BLSTM achieves an accuracy of over 99.1% across the China Physiological Signal Challenge 2018 and MIT-BIH arrhythmia databases (MADB), demonstrating great potential in lightweight devices.

Index Terms—Electrocardiogram (ECG), position perception circular convolution (PerC), Transformer.

I. INTRODUCTION

ARRHYTHMIA is a kind of common cardiovascular disease that is the primary factor for over 32% of all deaths worldwide [1]. As the most frequent arrhythmias, left bundle branch block (LBBB) and right bundle branch block (RBBB) are easily overlooked in clinical owing to their highly similar electrocardiogram (ECG) features. Unfortunately, healthy individuals may generate RBBB and this in turn induces LBBB;

the latter is the major cause of several complications such as heart failure and myocardial infarction [2], [3]. The prevalence of both gradually rises with age [4]. Consequently, an accurate and timely recognition of RBBB and LBBB has played an essential role in rendering the early warning and subsequently curing both diseases.

Recently, ECG has been a primary tool for clinicians to diagnose and distinguish various arrhythmias such as RBBB and LBBB. Visually inspecting ECG with the naked eye is currently the most traditional diagnostic pattern [5]. However, given several obstacles including strenuous manual inspection, inevitable human error, and explosive growth of data volumes, it is challenging to further promote traditional diagnostic strategy for the current clinical environment [6], [7]. Hence, an efficient automatic recognition strategy is desired and has become a beneficial endeavor for RBBB and LBBB detection.

Recent years have seen dramatic advances in automatic detection strategies for ECG recognition and they mostly resort to general machine learning (ML) algorithms. The artificial construction of feature extraction holds a dominant position for algorithm performance [8], [9]. Researchers often investigate feature extraction procedure with the help of various strategies or theories such as morphology [10], statistics [11], and wavelet transform [12]. However, applying conventional ML algorithms to ECG signal recognition is challenging due to the difficulties in inferior generalization ability, high-frequency occurrence of overfitting issue and the absence of standardized guidelines for feature extraction. These issues are increasingly impeding their utilization in the actual implementation of many application scenarios.

Since ResNet [13], deep convolutional neural networks (DCNNs) have dominated across application domains such as target detection [14] and ECG signals recognition [15]. With this trend, Researchers have shifted from designing handcrafted features to engineering DCNN models. Due to its self-learning capacity, DCNN can reach an automatic feature extraction procedure and thus has good generalization ability even when facing several categories of data sources. Although there have been various pure CNN models in recent years, their performances have been challenged by Transformer-like models due to the limitations of local receptive field [16], [17], [18], [19]. Furthermore, as previously stated in foundational

Received 31 March 2025; revised 3 September 2025; accepted 10 February 2026. Date of publication 4 March 2026; date of current version 2 September 2026. This work was supported in part by the Anhui Provincial Department of Education Project under Grant 2025AHGXZK40256, in part by the Quality Engineering Project of Bengbu Medical University under Grant 2025jyxm39, in part by the Industry-University Cooperation Horizontal Project titled “Development of Intelligent Warning Model for Persistent Ventricular Tachycardia,” in part by “512” Outstanding Talents Fostering Project of Bengbu Medical College under Grant BY51201312, and in part by the Natural Science Youth Project of Bengbu Medical University under Grant 2024byzd133. (Corresponding author: Jibin Wang.)

The authors are with the Anhui Key Laboratory of Digital Medicine and Intelligent Health, School of Medical Imaging, Bengbu Medical University, Bengbu 233030 China (e-mail: wangjibin0503@126.com; shibo@bbmu.edu.cn).

Digital Object Identifier 10.1109/TNNLS.2026.3665367

2162-237X © 2026 IEEE. All rights reserved, including rights for text and data mining, and training of artificial intelligence and similar technologies. Personal use is permitted, but republication/redistribution requires IEEE permission.

See <https://www.ieee.org/publications/rights/index.html> for more information.

Authorized licensed use limited to: China Agricultural University. Downloaded on September 09, 2026 at 12:44:30 UTC from IEEE Xplore. Restrictions apply.

work MetaFormer [20] that existing mainstream Transformer can be abstracted as a universal MetaFormer and empirical experiments have confirmed that the success of Transformer-like model is largely attributed to the general architecture MetaFormer.

To this end, we inherit the theoretical basis of MetaFormer and construct a novel MetaFormer-like model that is composed of both position perception circular convolution (PerC) [21] module and simple attention mechanism based on bidirectional long short-term memory (SimAM-BLSTM) known as PerC-SimAM-BLSTM [22]. PerC-SimAM-BLSTM boasts a global receptive field to generate location-sensitive representations achieving the extraction of local and global features and also actualizing interactive information between features. Ablation studies across the China Physiological Signal Challenge 2018 Database (CPSCDB) and MIT-BIH Arrhythmia Database (MADB) will assess the effectiveness of our proposed PerC-SimAM-BLSTM.

Our contribution contains several points.

- 1) To overcome the limitations of traditional convolutional receptive field, we present a novel MetaFormer-like model called PerC-SimAM-BLSTM with a global receptive field to generate location-sensitive representations. This model not only inherits the global feature extraction ability of Transformer but also enhances global-local feature acquisition ability of PerC.
- 2) To further improve the local feature extraction ability of PerC-SimAM-BLSTM, we construct a SimAM-BLSTM module to actualize interactive information between features.
- 3) We employ the proposed PerC-SimAM-BLSTM on RBBB and LBBB detection with two public datasets while obtaining better performance compared to the existing strategies. Our PerC-SimAM-BLSTM also strikes a better balance between performance and complexity from an empirical experiment perspective.
- 4) To our knowledge, this is the first attempt to incorporate the advantages of DCNN and Transformer to develop a pure CNN module for ECG signal recognition. Mathematical convergence proof ensures stable convergence of model training to the theoretical optimal solution from a theoretical perspective.

The remainder of this work is organized as follows. Section II reviews the recent research on RBBB and LBBB detection. Section III expounds the particulars of ECG database, PerC-SimAM-BLSTM, and evaluation framework to assess the effectiveness of PerC-SimAM-BLSTM. Section IV reveals the experimental results, including RBBB and LBBB recognition, model convergence, and detail analysis. Theoretical proof, complexity analysis, performance comparisons with state-of-the-art strategies, and limitations are given in Section V. Section VI summarizes this work.

II. RECENT WORK

Recent research about RBBB and LBBB detection can be grouped into general ML algorithms and DCNN algorithms.

- 1) *General ML algorithm*: Sahoo et al. [23] investigated a system mechanism with multiresolution wavelet features

and a support vector machine (SVM) classifier for RBBB and LBBB signals identification. Shi et al. [24] developed an ML algorithm based on several region features with an ensemble classifier for normal sinus rhythm (NSR), RBBB, and LBBB detection. Tuncer et al. [25] designed an intelligence-based classification methodology with hexadecimal local pattern (HLP) mechanism and one nearest neighborhood (ONN) classifier for RBBB and LBBB fragments. Jha and Kolekar [26] investigated a robust method to utilize Q-wavelet transform features and SVM to distinguish RBBB and LBBB. Chandrasekar et al. [27] applied an ML system mechanism that combines peak characteristics and a quadratic SVM classifier for RBBB and LBBB detection. Besides, Nahak et al. [28] implemented an ECG detection method based on wavelet scattering transform (WST) and the SVM classifier for the identification of RBBB and LBBB episodes.

While conventional ML methods (e.g., wavelet features+SVM [23], HLP+ONN [25], and Q-wavelet+SVM [26]) have demonstrated utility in detecting BBB, they suffer from three critical constraints. First, these methods rely heavily on handcrafted feature engineering, which demands domain expertise and lacks standardized guidelines, leading to suboptimal generalizability across diverse ECG datasets. Second, the shallow classifiers struggle to model the complex spatiotemporal dependencies in ECG, resulting in limited sensitivity to subtle morphological variations in LBBB and RBBB. Third, manual feature extraction is susceptible to noise and interpatient variability often causing overfitting in small-scale datasets. These issues collectively hinder their robustness in clinical deployment.

- 1) *DCNN Algorithm*: Matamoros et al. [29] investigated a DCNN architecture with the utilization of bidirectional recurrent neural network (BRNN) to categorize RBBB and LBBB. Bouny et al. [30] proposed a novel wavelet CNN (MA-CNN) framework with multiple wavelet features to implement RBBB and LBBB analysis. Yao et al. [31] developed a methodology based on attention mechanism and LSTM for a variety of ECG signal recognition. Faust and Urabc [32] constructed a ResNet framework and several RR interval features to distinguish many types of heartbeats. Singh and Sharma [33] presented a DCNN model employing several ResNet modules and gradient-weighted class activation mapping strategy for RBBB and LBBB detection. Zhang et al. [34] constructed a CNN-based intelligent recognition mechanism with attention U-net for ECG signal analysis. Zahid et al. [35] developed a DCNN classification model with self-operational network for the classification of RBBB and LBBB. Din et al. [36] proposed a comprehensive feature fusion framework with integrating CNN, LSTM, and Transformer for RBBB and LBBB signals identification. Recently, Cheng et al. [37] put forward a lead feature guide network (LFG-Net) for the classification of RBBB and LBBB episodes.

TABLE I
SPECIFICS OF ECG EPISODES IN THIS WORK

Database	Sort	Quantity
China Physiological Signal Challenge 2018 database (CPSCDB)	NSR	9773
	LBBB	4802
	RBBB	8195
MIT-BIH arrhythmia database (MADB)	NSR	7581
	LBBB	3368
	RBBB	4350

Recent DCNN-based approaches (e.g., ResNet [33], CNN+U-Net [34], and self-operational networks [35]) mitigate manual feature design but face inherent architectural limitations. Pure CNNs exhibit restricted local receptive fields, failing to capture global context. Though Transformer-integrating frameworks enhance global modeling, their self-attention mechanisms incur high computational overhead, complicating deployment on resource-constrained devices. Moreover, the existing hybrid models often inadequately balance local feature extraction and global dependency modeling, limiting discriminative power.

- 1) *Our Research*: To this end, we propose **PerC-SimAM-BLSTM**, which synergizes the global topology awareness of PerC and feature refinement capability of SimAM-BLSTM within a MetaFormer-like architecture achieving efficient global-local feature fusion.

III. METHODS AND MATERIALS

A. Database

In this study, ECG recordings of CPSCDB and MADB are selected to evaluate the validity of our proposed PerC-SimAM-BLSTM. Both databases are international public datasets, and the particulars about ECG database are described.

- 1) *CPSCDB*: This database comprises 6877 12-lead ECG recordings from 11 community hospitals, and each recording sampled via a frequency of 500 Hz lasts about 6–60 s [38].
- 2) *MADB*: This database comprises 48 half-hour 2-lead recordings of 47 patients, and each recording is sampled via a frequency of 360 Hz [39].

Fig. 1 also describes an established case of ECG episodes for LBBB, RBBB, and NSR types.

To reduce computational burden and retain data consistency, downsampling process is executed on ECG recordings from CPSCDB and MADB at 250 Hz and segmented into 2-s episodes. Furthermore, we take advantage of wavelet transform to filter the segmented episodes and Z-score standardization to the filtered ECG episodes for optimizing training [40], [41]. The specifics of ECG episodes in this study are displayed in Table I.

B. PerC-SimAM-BLSTM

As previously demonstrated in research MetaFormer [20] that the existing mainstream Transformer can be abstracted

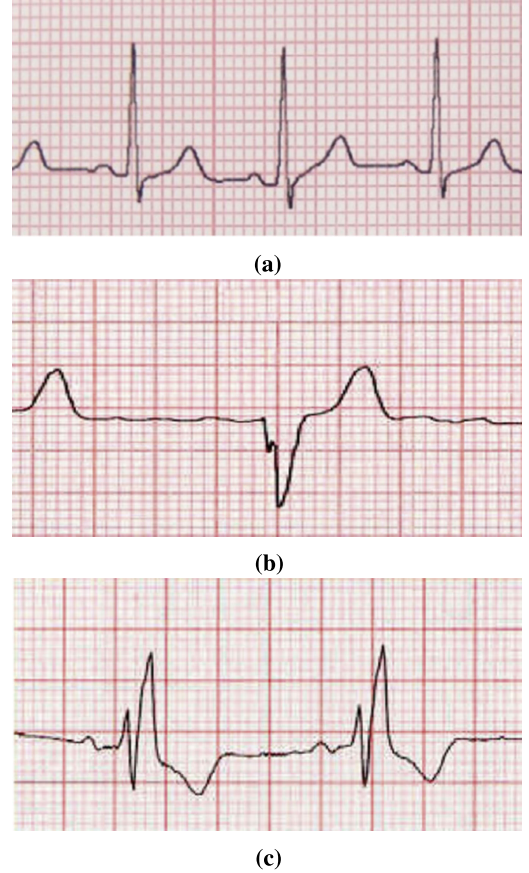


Fig. 1. Established case of ECG episodes for (a) LBBB, (b) RBBB, and (c) NSR.

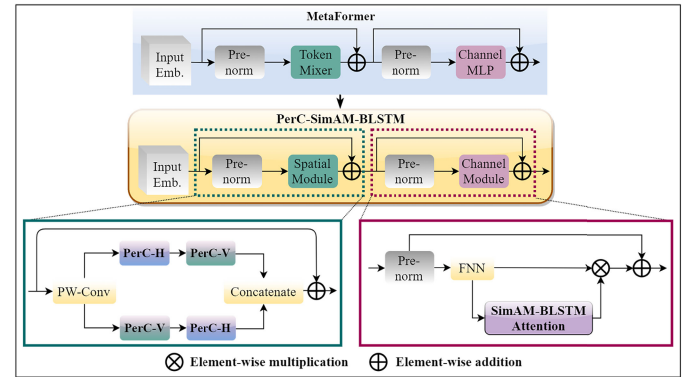


Fig. 2. Architecture overview of PerC-SimAM-BLSTM. Here Pre-norm: Pre-normalization, PW-Conv: Depth-wise convolution, PerC-V: PerC on vertical orientation, PerC-H: PerC across horizontal orientation, and FNN: feature forward network.

as a universal MetaFormer and empirical experiments have confirmed that the success of Transformer/MLP-like model is largely attributed to the general architecture MetaFormer instead of the equipped specific token mixer. Based on this, we inherit the idea of Transformer and its variants such as Vision Transformer (ViT) [17], Swin Transformer (SwinT) [18], and MLP-Mixer [19], develop PerC module as a new token mixer, and construct the SimAM-BLSTM module as a new Channel MLP to compensate for feature discrimination ability. Fig. 2 illustrates the overview and research idea of our PerC-SimAM-BLSTM.

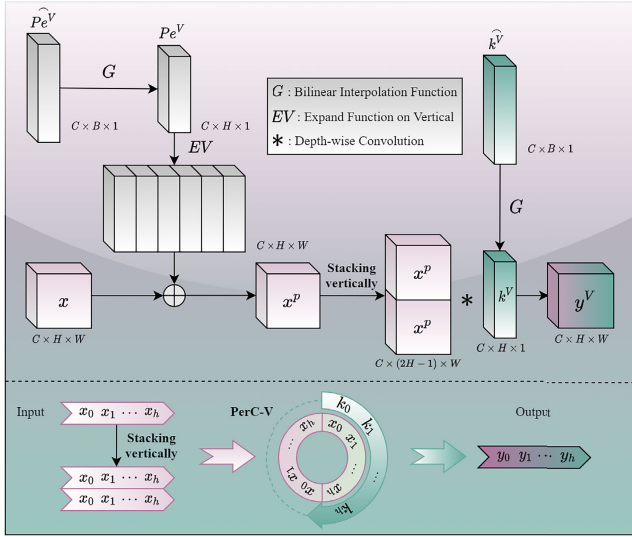


Fig. 3. Architecture overview of PerC on vertical orientation.

Traditional convolutions extract local features within fixed kernel windows, limiting their ability to model long-range dependencies in ECG signals. PerC overcomes this via circular convolution combined with adaptive position encoding. Specifically, it includes several advantages.

- 1) *Circular Convolution*: Unlike standard convolution, PerC treats input dimensions as circular buffers. For vertical orientation (PerC-V), it connects the first and last rows of the input matrix; for horizontal orientation (PerC-H), it links the first and last columns. This enables global receptive fields by allowing information to flow cyclically across boundaries, capturing periodic patterns in ECG.
- 2) *Position Perception*: PerC injects learnable position embeddings into the input. These embeddings are generated via bilinear interpolation and expanded along the target orientation. They encode absolute spatial-temporal locations, ensuring the model discerns subtle positional variations.
- 3) *Joint Effect*: The instance-specific kernels dynamically weight features based on the position-enhanced inputs allowing PerC to extract location-sensitive global features while retaining local detail. This synergizes with SimAM-BLSTM to model both structural and contextual dependencies in arrhythmias. Imagine overlaying a grid on ECG feature map. PerC-V scans each column as a "circular tube," while PerC-H scans each row as a "ring." This ensures that every pixel perceives global positional context, crucial for identifying spatially distributed arrhythmia features. PerC synergizes local feature extraction with global topology awareness, bridging CNN efficiency and Transformer expressiveness.

1) *Position Perception Circular Convolution (PerC)*: As illustrated in Fig. 2, our proposed PerC has two patterns, one is PerC of PerC-V and the other is PerC of PerC-H. These settings make all pixels with the same column and same row covered by the receptive fields of both PerC-V and PerC-H,

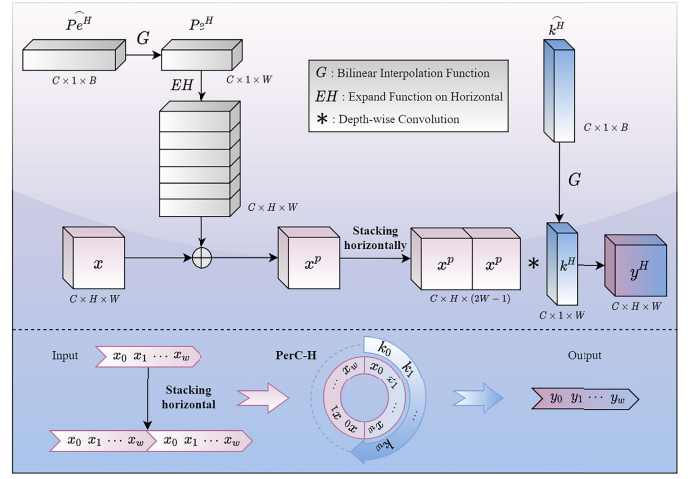


Fig. 4. Architecture overview of PerC on horizontal orientation.

respectively. Jointly leveraging PerC-V and PerC-H patterns can extract global features of the inputs. Specifically, we assume that the input $x \in \mathbb{R}^{1 \times H \times W}$ and the output of PerC-V at location (i, j) can be described

$$\begin{cases} \text{Pe}^V = G(\widehat{\text{Pe}}^V) = [\text{Pe}_0^V, \text{Pe}_1^V, \dots, \text{Pe}_{H-1}^V]^T \\ \text{Pe}_e^V = \text{EV}(\text{Pe}^V, \theta) \\ k^V = G(\widehat{k}^V) = [k_0^V, k_1^V, \dots, k_{H-1}^V] \\ x^p = x + \text{Pe}_e^V \\ y_{i,j}^V = \sum_{t \in (0, H-1)} (k)_t^V x_{[(i+t) \bmod H, j]}^p \end{cases} \quad (1)$$

where Pe^V indicates the position embedding (PE) and can be computed by base embedding $\widehat{\text{Pe}}^V$ via a bilinear interpolation function $G(\cdot)$. Pe_e^V is mapped PE and k^V represents instance kernel. $\text{EV}(\cdot)$ is an expand function of vertical orientation. By duplicating the input θ times, $\text{EV}(\cdot)$ concatenates these copied vectors along horizontal orientation to generate a $H \times W$ -sized PE matrix. Fig. 3 provides the details of calculation process in an instance where the model input is a 1-D vector on vertical orientation.

Similarly, we can compute the output of PerC-H at location (i, j)

$$\begin{cases} \text{Pe}^H = G(\widehat{\text{Pe}}^H) = [\text{Pe}_0^H, \text{Pe}_1^H, \dots, \text{Pe}_{W-1}^H]^T \\ \text{Pe}_e^H = \text{EH}(\text{Pe}^H, \theta) \\ k^H = G(\widehat{k}^H) = [k_0^H, k_1^H, \dots, k_{W-1}^H] \\ x^p = x + \text{Pe}_e^H \\ y_{i,j}^H = \sum_{t \in (0, W-1)} (k)_t^H x_{[i, (j+t) \bmod W]}^p \end{cases} \quad (2)$$

where $\text{EH}(\cdot)$ is the input vector along horizontal orientation. Fig. 4 gives the details for calculation process in an instance that model input is a 1-D vector on horizontal orientation. Therefore, our proposed convolution is referred to as circular convolution with a global receptive field to extract global features.

2) *SimAM-BLSTM Module*: To further extract local features, we put forward SimAM-BLSTM to couple the global features stemmed from PerCNet. It is well known that both hidden states of forward $\vec{h}_t$ and backward $\overleftarrow{h}_t$ directions in BLSTM are simply concatenated into output $V_t = [\vec{h}_t; \overleftarrow{h}_t]$, which results in limited performance. Given that the existing attention mechanisms only generate 1-D or 2-D importance weights, this limits the network's feature extraction ability [22], [42]. For this reason, we develop a simple and effective SimAM-BLSTM refinement module to generate more abundant 3-D weights. Visual neuroscience theory indicates that assigning higher priority to neurons with clear spatial suppression effects can be achieved by measuring the linear separability between one target neuron and other neurons [43]. According to the neuroscience findings, the energy function for each neuron is defined to calculate 3-D weights

$$e_a(\omega_a, b_a, y, x_i) = (y_a - \hat{a})^2 + \frac{1}{N-1} \sum_{i=1}^N (y_o - \hat{x}_i)^2 \quad (3)$$

where $\hat{a}, \hat{x}_i$ is the linear transforms of a and x_i . a and x_i are the target neuron and other neurons in a single channel of the inputs $X \in R^{C \times H \times W}$, i is the index over spatial dimension and $N = H \times W$ is the number of neurons on the corresponding channel. ω_a and b_a are weight and bias parameters, respectively. Equation (3) aims to attain the minimal value when $\hat{a}$ equals to y_a and all other $\hat{x}_i$ are y_o . For simplicity, we use binary labels for y_a and y_o and merge a regularizer into (3) and the energy function can be further deformed

$$e_a(\omega_a, b_a, y, x_i) = \frac{1}{N-1} \sum_{i=1}^N [-1 - (\omega_a(x)_i + b_a)]^2 + [1 - (\omega_a a + b_a)]^2 + \eta \omega_a^2, \quad (4)$$

luckily, (4) has a fast closed-form solution for ω_a, b_a and can be easily acquired

$$\begin{cases} \omega_a = -\frac{2(a - \mu_a)}{(a - \mu_a)^2 + 2\sigma_a^2 + 2\eta} \\ b_a = -\frac{1}{2}(a + \mu_a)\omega_a \end{cases} \quad (5)$$

where $\mu_a = \frac{1}{N-1} \sum_{i=1}^{N-1} x_i$ and $\sigma_a^2 = \frac{1}{N-2} \sum_{i=1}^{N-1} (x_i - \mu_a)^2$ are mean and variance over all neurons except a in that channel, respectively. Accordingly, we can compute the minimal energy

$$e_a^* = \frac{4(\hat{\sigma}^2 + \eta)}{(a - \hat{\mu})^2 + 2\hat{\sigma}^2 + 2\eta} \quad (6)$$

where $\hat{\mu} = \frac{1}{N} \sum_{i=1}^N x_i$ and $\hat{\sigma}^2 = \frac{1}{N-1} \sum_{i=1}^{N-1} (x_i - \hat{\mu})^2$.

Equation (6) indicates that the lower energy e_a^* , the neuron a is more discriminative compared to nearby neurons and more essential for visual processing. Hence, the importance of each neuron can be achieved via $\frac{1}{e_a^*}$. According to [43], attention modulation in the human brain generally implements as a gain effect on neuronal responses. Simultaneously integrating with

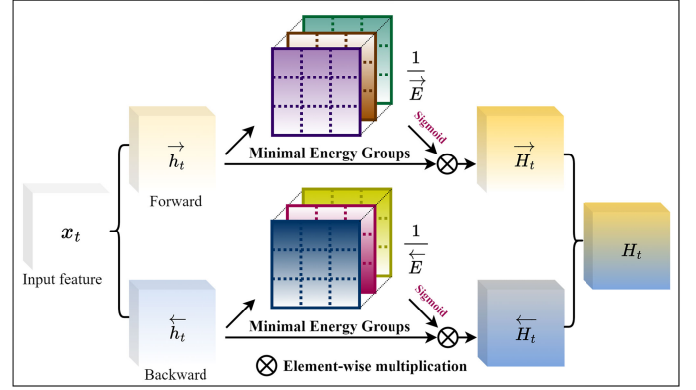


Fig. 5. Architecture overview of SimAM-BLSTM module.

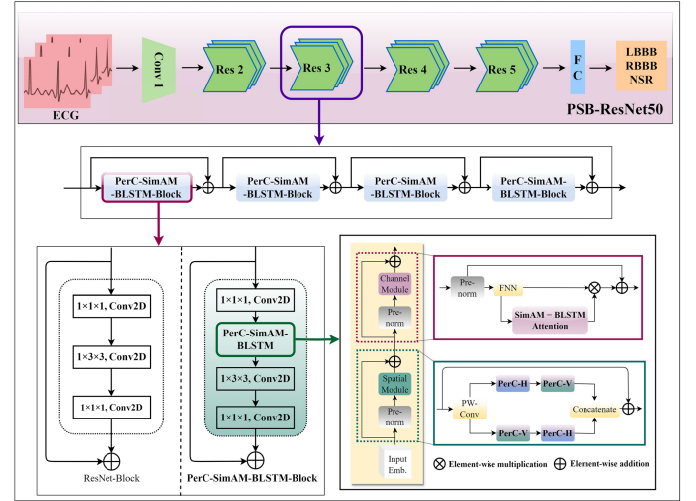


Fig. 6. Architecture overview of PSB-ResNet50.

BLSTM in Fig. 5, we utilize a scaling operator for feature refinement

$$\begin{cases} \vec{H}_t = \text{sigmoid}\left(\frac{1}{\vec{E}}\right) \vec{h}_t \\ \overleftarrow{H}_t = \text{sigmoid}\left(\frac{1}{\overleftarrow{E}}\right) \overleftarrow{h}_t \\ H_t = [\vec{H}_t; \overleftarrow{H}_t] \end{cases} \quad (7)$$

where $\vec{E}$ and $\overleftarrow{E}$ groups all e_a^* across channel and spatial dimensions of both directions. Our proposed module is inspired from neuroscience theory and simple to implement and employ along with existing architectures.

C. Architecture Details

In this section, we aim to confirm the effectiveness of PerC-SimAM-BLSTM module and implant it into ResNet50 known as PSB-ResNet50. The detailed architecture configuration of PSB-ResNet50 is illustrated in Fig. 6. PSB-ResNet50 is built upon 5 Res building blocks on top of each other, and any Res building block is composed of four PerC-SimAM-BLSTM blocks. The purpose of Softmax function is to compute the identification probability for three types of heartbeats (LBBB, RBBB, or NSR) and the Adam optimizer is also leveraged to

TABLE II

DETECTION PERFORMANCE RESULTS OF FIVE PROPOSED ARCHITECTURES ACROSS CPSCDB. HERE, LSTM IN PSL-ResNet50 REPLACES BLSTM IN PSB-ResNet50

Architecture	Category	ACC(%)	SEN(%)	SPF(%)
MLP-Mixer-ResNet50 [19]	NSR	99.0	98.4	99.3
	LBBB	99.1	98.5	99.2
	RBBB	99.1	98.5	99.0
ViT-ResNet50 [17]	NSR	99.1	98.7	99.2
	LBBB	99.2	98.6	99.3
	RBBB	99.1	98.6	99.3
SwinT-ResNet50 [18]	NSR	99.2	98.8	99.3
	LBBB	99.3	98.7	99.4
	RBBB	99.2	98.8	99.4
PSL-ResNet50	NSR	99.5	99.0	99.3
	LBBB	99.5	98.9	99.4
	RBBB	99.3	98.9	99.5
PSB-ResNet50	NSR	99.5	99.1	99.3
	LBBB	99.4	98.9	99.5
	RBBB	99.3	99.0	99.5

optimize the proposed PSB-ResNet50 with a mini-batch size of 64. The algorithm pseudocode is shown in Algorithm 1.

IV. RESULTS

In this section, we conduct a series of empirical experiments with a total of 22770 ECG episodes involving 9773 NSR, 4802 LBBB, and 8195 RBBB on CPSCDB, as well as 15299 ECG episodes involving 7581 NSR, 3368 LBBB, and 4350 RBBB on MADB. To confirm the validity of PerC-SimAM-BLSTM, ViT [17], SwinT [18], and MLP-Mixer [19] are used to construct ViT-ResNet50, SwinT-ResNet50, and MLP-Mixer-ResNet50, and the models are designed via the same compositions as PSB-ResNet50, except that PerC-SimAM-BLSTM is replaced with ViT, SwinT, and MLP-Mixer. Also, accuracy (ACC), sensitivity (SEN), and specificity (SPF) are regarded as evaluation indicators.

A. Signal Detection

Table II illustrates the detection results of five proposed architectures across CPSCDB; 99.1% of ECG episodes can be accurately distinguished as NSR episodes via PSB-ResNet50, 98.9% of ECG episodes can be accurately distinguished as LBBB episodes, and 1.0% of RBBB episodes are falsely distinguished as other categories. In terms of MLP-Mixer-ResNet50, ViT-ResNet50, SwinT-ResNet50, and PSL-ResNet50, 98.4%, 98.7%, 98.8%, and 99.0% of ECG episodes are exactly distinguished as NSR episodes, respectively; 98.5%, 98.6%, 98.7%, and 98.9% of ECG episodes are accurately distinguished as LBBB episodes, respectively; and 1.5%, 1.4%, 1.2%, and 1.1% of RBBB episodes are falsely distinguished as other categories, respectively.

Table III illustrates the detection results of five proposed architectures across MADB; 98.7% of ECG episodes are accurately distinguished as NSR episodes via PSB-ResNet50, 98.8% of ECG episodes can be accurately distinguished as LBBB episodes, and 1.3% of RBBB episodes are falsely distinguished as other categories. In terms of MLP-Mixer-ResNet50, ViT-ResNet50, SwinT-ResNet50, and PSL-ResNet50, 98.2%, 98.4%, 98.5%, and

Algorithm 1 PerC-SimAM-BLSTM Algorithm Pseudocode

```

function PERC_SIMAM_BLSTM(x)                                ▶Input
  PERC-V
  function PERC_VERTICAL(x)
    PeV ← BilinearInterpolation(PeV)                        ▶ Eq.(1)
    PeVe ← EV(PeV, θ)                                       ▶ Eq.(1)
    kV ← BilinearInterpolation(kbaseV)                     ▶ Eq.(1)
    xp ← x + PeVe                                           ▶ Eq.(1)
    yi,jV ←  $\sum_{t=0}^{H-1} k_t^V x_{[(i+t) \bmod H, j]}^p$            ▶ Eq.(1)
    return yV
  end function
  PERC-H
  function PERC_HORIZONTAL(x)
    PeH ← BilinearInterpolation(PeH)                        ▶ Eq.(2)
    PeHe ← EH(PeH, θ)                                       ▶ Eq.(2)
    kH ← BilinearInterpolation(kbaseH)                     ▶ Eq.(2)
    xp ← x + PeHe                                           ▶ Eq.(2)
    yi,jH ←  $\sum_{t=0}^{W-1} k_t^H x_{[i, (j+t) \bmod W]}^p$            ▶ Eq.(2)
    return yH
  end function
  SIMAM-BLSTM
  function SIMAM_BLSTM(X)
    function ENERGY(a, xi)
      μ ←  $\frac{1}{N-1} \sum x_i$                                      ▶ Eq.(3)-(4)
      σ2 ←  $\frac{1}{N-2} \sum (x_i - \mu)^2$ 
      ω ←  $\frac{-2(a-\mu)}{(a-\mu)^2 + 2\sigma^2 + 2\eta}$                        ▶ Eq.(5)
      e* ←  $\frac{4(\sigma^2 + \eta)}{(a-\mu)^2 + 2\sigma^2 + 2\eta}$                  ▶ Eq.(6)
      return 1/e*
    end function
    (H, H̄) ← BLSTM(X)
    Ē ← Energy(H̄, neighbors(H̄))
    Ē ← Energy(H̄, neighbors(H̄))
    H̄t ← sigmoid(1/Ē) ⊙ h̄t                                ▶ Eq.(7)
    H̄t ← sigmoid(1/Ē) ⊙ h̄t                                ▶ Eq.(7)
    return [H̄t; H̄t]                                       ▶ Eq.(7)
  end function
  MAIN PROCESS
  yV ← PerC_Vertical(x)
  yH ← PerC_Horizontal(x)
  y ← yV + yH
  y ← SimAM_BLSTM(y)
  return y + x
end function
function PSB-RESNET50(ECG)
  x ← WaveletFilter(Z_Score(Downsample(ECG)))
  for 5 ResBlock do
    x ← ResBlock(x, [PerC_SimAM_BLSTM]4)
  end for
  logits ← GlobalAvgPool(FC(x))
  return Softmax(logits)                                ▶LBBB/RBBB/NSR
end function

```

98.6% of ECG episodes are exactly distinguished as NSR episodes, respectively; 98.1%, 98.2%, 98.6%, and 98.7% of ECG episodes can be accurately distinguished as LBBB

TABLE III
DETECTION PERFORMANCE RESULTS OF FIVE PROPOSED
ARCHITECTURES ACROSS MADB

Architecture	Category	ACC(%)	SEN(%)	SPF(%)
MLP-Mixer-ResNet50 [19]	NSR	98.6	98.2	99.1
	LBBB	98.9	98.1	99.1
	RBBB	98.8	98.2	99.0
ViT-ResNet50 [17]	NSR	98.8	98.4	99.1
	LBBB	99.0	98.2	99.2
	RBBB	98.9	98.3	99.2
SwinT-ResNet50 [18]	NSR	98.9	98.5	99.2
	LBBB	99.1	98.6	99.2
	RBBB	99.0	98.4	99.3
PSL-ResNet50	NSR	99.0	98.6	99.4
	LBBB	99.2	98.7	99.2
	RBBB	99.1	98.7	99.3
PSB-ResNet50	NSR	99.0	98.7	99.3
	LBBB	99.2	98.8	99.3
	RBBB	99.2	98.7	99.4

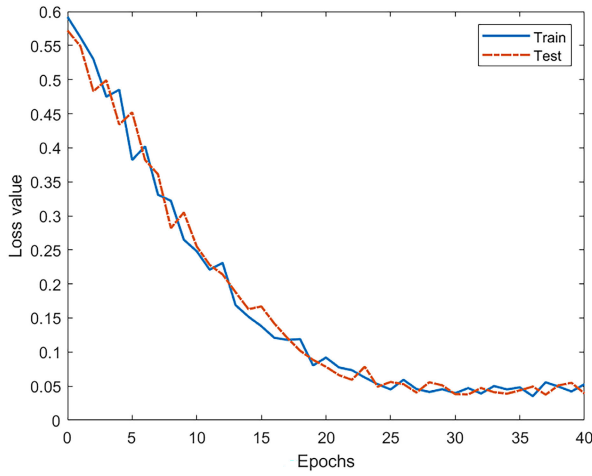


Fig. 7. Training curve tendency for PSB-ResNet50 across CPSCDB.

episodes; and 1.8%, 1.7%, 1.6%, and 1.3% of RBBB episodes are falsely distinguished as other categories, respectively.

Tables II and III indicate that PSB-ResNet50 achieves the best detection performance compared to four control group models across both databases. Evaluation indicators are 0.1%–0.7% better than those of control group models confirming the effectiveness of PerC-SimAM-BLSTM and more outstanding discrimination ability than ViT, SwinT, and MLP-Mixer for LBBB and RBBB detection. Especially in PSB-ResNet50, PSL-ResNet50 integrated by replacing BLSTM with LSTM still achieves better performance than MLP-Mixer-ResNet50, ViT-ResNet50, and SwinT-ResNet50. This result confirms the high efficiency of PerC-SimAM-B/LSTM-class architecture again. The performance of PSL-ResNet50 is slightly inferior to that of PSB-ResNet50 because of bidirectional richer feature representations in BLSTM.

B. Model Convergence

The convergence performance of the network shows the robust generalization ability of network architecture. Figs. 7 and 8 illustrate the training curve tendency of PSB-ResNet50 across both databases. PSB-ResNet50

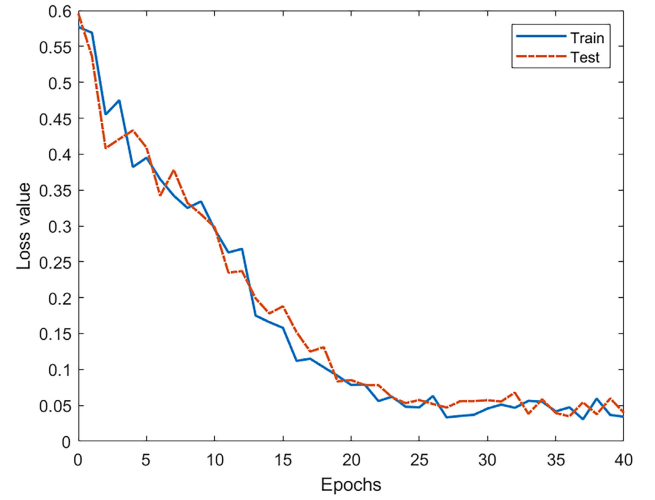


Fig. 8. Training curve tendency for PSB-ResNet50 across MADB.

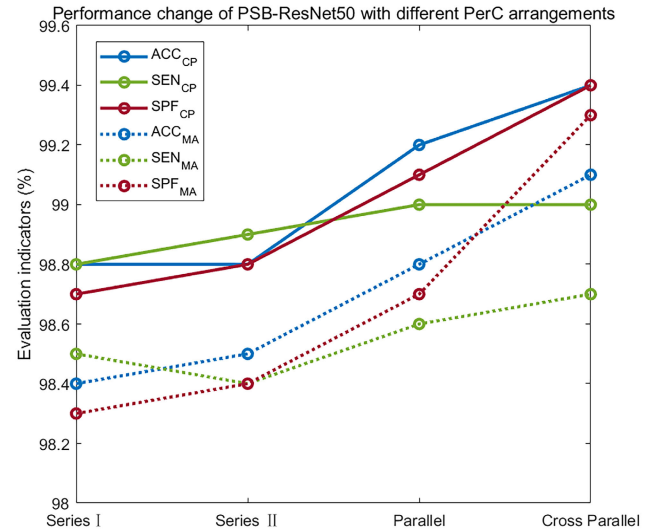


Fig. 9. Evaluation indicator change for PSB-ResNet50 with different PerC arrangement structures across both databases.

converges within approximately 22 and 23 iteration epochs during the training and testing phases across CPSCDB as well as about 20 and 21 iteration epochs during the train and test courses across MADB, respectively. The results confirm the reliable model robustness of our PerC-SimAM-BLSTM with different data sources.

C. Detailed Analysis

Tables IV and V indicate the overall detection performances for PSB-ResNet50 with different PerC arrangement structures within spatial module in Fig. 2 across both databases. PSB-ResNet50 achieves the best detection performance when PerC crosses the parallel strategy $\text{PerC-H} \propto \text{PerC-V} \parallel$ $\text{PerC-V} \propto \text{PerC-H}$ is utilized. We speculate that compared with series and parallel patterns, an essential reason for performance improvement is that this strategy can achieve interactive

TABLE IV

OVERALL DETECTION PERFORMANCE FOR PSB-RESNET50 WITH DIFFERENT PERC ARRANGEMENT STRUCTURES WITHIN SPATIAL MODULE IN FIG. 2 ACROSS CPSCDB. HERE, $\propto$ REPRESENTS SERIES CONNECTION AND $\parallel$ REPRESENTS PARALLEL CONNECTION

Arrangement structure	ACC(%)	SEN(%)	SPF(%)
$PerC - H \propto PerC - H \propto PerC - V \propto PerC - V$	98.8	98.8	98.7
$PerC - V \propto PerC - V \propto PerC - H \propto PerC - H$	98.8	98.9	98.8
$PerC - H \propto PerC - H \parallel PerC - V \propto PerC - V$	99.2	99.0	99.1
$PerC - H \propto PerC - V \parallel PerC - V \propto PerC - H(Ours)$	99.4	99.0	99.4

TABLE V

OVERALL DETECTION PERFORMANCE FOR PSB-RESNET50 WITH DIFFERENT PERC ARRANGEMENT STRUCTURES WITHIN SPATIAL MODULE IN FIG. 2 ACROSS MADB

Arrangement structure	ACC(%)	SEN(%)	SPF(%)
$PerC - H \propto PerC - H \propto PerC - V \propto PerC - V$	98.4	98.5	98.3
$PerC - V \propto PerC - V \propto PerC - H \propto PerC - H$	98.5	98.4	98.4
$PerC - H \propto PerC - H \parallel PerC - V \propto PerC - V$	98.8	98.6	98.7
$PerC - H \propto PerC - V \parallel PerC - V \propto PerC - H(Ours)$	99.1	98.7	99.3

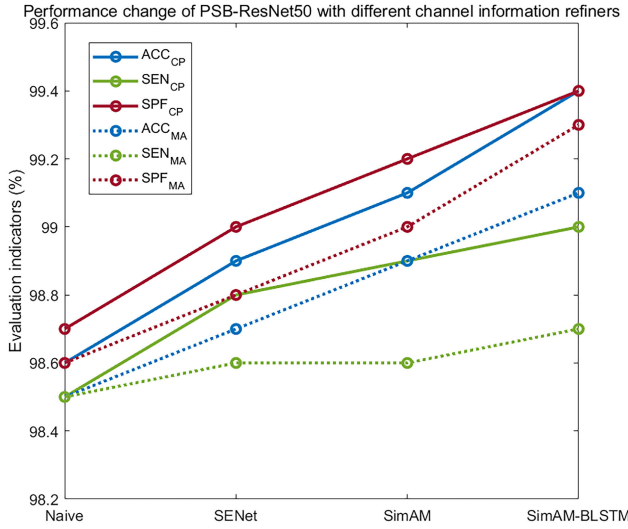


Fig. 10. Evaluation indicator change for PSB-ResNet50 with different channel information refiners across both databases.

TABLE VI

OVERALL DETECTION PERFORMANCE FOR PSB-RESNET50 WITH DIFFERENT CHANNEL INFORMATION REFINERS WITHIN CHANNEL MODULE IN FIG. 2 ACROSS CPSCDB

Channel information refiner	ACC(%)	SEN(%)	SPF(%)
Naive version	98.6	98.5	98.7
SENet [44]	98.9	98.8	99.0
SimAM module [22]	99.1	98.9	99.2
<i>SimAM - BLSTM(Ours)</i>	99.4	99.0	99.4

information between local and global feature modules. Fig. 9 displays the change trend of evaluation indicators for PSB-ResNet50 with different PerC arrangement structures across both databases authenticating the conclusion again.

Tables VI and VII show the overall detection performances for PSB-ResNet50 with different channel information refiners within spatial module in Fig. 2 across both databases. PSB-ResNet50 achieves the best performance when SimAM-BLSTM is integrated into PerC-SimAM-BLSTM.

TABLE VII

OVERALL DETECTION PERFORMANCE FOR PSB-RESNET50 WITH DIFFERENT CHANNEL INFORMATION REFINERS WITHIN CHANNEL MODULE IN FIG. 2 ACROSS MADB

Channel information refiner	ACC(%)	SEN(%)	SPF(%)
Naive version	98.5	98.5	98.6
SENet [44]	98.7	98.6	98.8
SimAM module [22]	98.9	98.6	99.0
<i>SimAM - BLSTM(Ours)</i>	99.1	98.7	99.3

The principal reason is attributed to SimAM-BLSTM not only leveraging rich bidirectional feature maps in BLSTM but also inheriting superior feature acquisition ability of SimAM module. Fig. 10 illustrates the change trend of evaluation indicators for PSB-ResNet50 with different channel information refiners across both databases authenticating the conclusion again.

V. DISCUSSION

A. Theoretical Proof

Let the training dataset be $\mathcal{D} = \{(x_i, y_i)\}_{i=1}^N$, where $x_i \in \mathbb{R}^{H \times W}$ is an ECG signal segment and $y_i \in \{0, 1\}^3$ is the category label (LBBB, RBBB, and NSR). The model parameters θ and the cross-entropy loss are denoted

$$\mathcal{L}(\theta) = -\frac{1}{N} \sum_{i=1}^N \sum_{c=1}^3 y_{i,c} \log p_{i,c}(\theta) \quad (8)$$

where $p_{i,c}(\theta) = \text{Softmax}(\text{PSB-ResNet50}(x_i; \theta))_c$ is the predicted probability. The optimization objective is to minimize $\mathcal{L}(\theta)$ with Adam optimizer. The following assumptions for convergence (Assumptions 1–3) must hold.

- 1) *Lipschitz Continuity*: There exists a constant $L > 0$ such that: $\|\nabla_{\theta} \mathcal{L}(\theta_1) - \nabla_{\theta} \mathcal{L}(\theta_2)\| \leq L \|\theta_1 - \theta_2\|$.
- 2) *Bounded Gradient*: There exists a constant $G > 0$ such that $\|\nabla_{\theta} \mathcal{L}(\theta)\| \leq G$.
- 3) *Adam Learning Rate Conditions*: The learning rate η_t satisfies: $\sum_{t=1}^{\infty} \eta_t = \infty$, $\sum_{t=1}^{\infty} \eta_t^2 < \infty$.

The vertical convolution output of PerC is

$$y_{i,j}^V = \sum_{t=0}^{H-1} k_t^V x_{[(i+t) \bmod H, j]}^p \quad (9)$$

where $x^p = x + \text{Pe}_e^V$. The position encoding Pe_e^V is generated via bilinear interpolation $G(\cdot)$ and normalized. Due to the cyclic boundary property of circular convolution, its gradient is

$$\frac{\partial \mathcal{L}}{\partial k_t^V} = \sum_{i,j} \frac{\partial \mathcal{L}}{\partial y_{i,j}^V} x_{[(i+t) \bmod H, j]}^p \quad (10)$$

Since x^p and k_t^V are normalized, gradient $\partial \mathcal{L} / \partial k_t^V$ is bounded, thereby avoiding explosion or vanishing. $\text{Sigmoid}(1/e_a^*) \in (0, 1)$, its gradient $\partial s / \partial e_a^*$ is bounded by the sigmoid derivative upper bound $\frac{1}{4}$

$$\left\| \frac{\partial s}{\partial e_a^*} \right\| \leq \frac{1}{4}. \quad (11)$$

TABLE VIII

COMPLEXITY COMPARISON OF OUR PROPOSED PERC-SIMAM-BLSTM WITH SEVERAL CLASSICAL STRATEGIES ACROSS CPSCDB. THE MODELS LISTED IN THE TABLE ARE NOT SINGLE MODELS BUT BASE MODELS AND THE SPECIFIC EXPERIMENTAL MODELS ARE STACKED ON TOP OF THESE BASE MODELS

Frameworks	Parameters (M)	FLOPs (GMac)	ACC(%)
ResNet [33]	1.10	0.89	97.9
LSTM [31]	0.97	1.12	96.3
U-Net [34]	2.92	2.57	94.8
Transformer [36]	4.30	3.10	99.6
SimAM [22]	3.20	1.72	99.1
<i>PerC-SimAM-BLSTM(Ours)</i>	2.43	1.96	99.4

The bidirectional propagation of BLSTM controlled by gating mechanisms (input gate i_t and forget gate f_t) updates parameters via

$$\frac{\partial \mathcal{L}}{\partial W_{\text{LSTM}}} = \sum_i \frac{\partial \mathcal{L}}{\partial h_t} \cdot \frac{\partial h_t}{\partial W_{\text{LSTM}}}. \quad (12)$$

LSTM's gradient clipping can further guarantee stable gradients $\partial h_t / \partial W_{\text{LSTM}}$. ResNet block satisfies: $y = \mathcal{F}(x, \{W_i\}) + x$ with gradient: $\frac{\partial \mathcal{L}}{\partial x} = \frac{\partial \mathcal{L}}{\partial y} \cdot (1 + \frac{\partial \mathcal{F}}{\partial x})$. Here, the identity mapping term 1 ensures linear gradient propagation, avoiding vanishing gradients in deep networks.

Theorem 1: Under Assumptions 1–3, the parameter sequence $\{\theta_t\}$ of PerC-SimAM-BLSTM satisfies

$$\lim_{t \rightarrow \infty} \nabla_{\theta} \mathcal{L}(\theta_t) = 0, \quad (13)$$

i.e., the gradient of loss function almost surely converges to zero and the model converges to a stationary point.

Proof:

- 1) *Bounded Gradient:* From module-level analysis

$$\|\nabla_{\theta} \mathcal{L}(\theta)\| \leq G. \quad (14)$$

- 2) *Adam Optimizer Convergence:* With learning rates satisfying $\sum \eta_t = \infty$, $\sum \eta_t^2 < \infty$, Adam's update rule

$$\theta_{t+1} = \theta_t - \eta_t \cdot \frac{\hat{m}_t}{\sqrt{\hat{v}_t} + \epsilon} \quad (15)$$

where $\hat{m}_t$ and $\hat{v}_t$ are bias-corrected momentum terms, which ensures

$$\sum_{t=1}^T \eta_t \mathbb{E} [\|\nabla_{\theta} \mathcal{L}(\theta_t)\|^2] \leq C, \quad (16)$$

for a constant C dependent on G, L, T condition. As $T \rightarrow \infty$, $\mathbb{E} [\|\nabla_{\theta} \mathcal{L}(\theta_t)\|^2] \rightarrow 0$.

- 3) *Empirical Validation:* Figs. 7 and 8 indicate that PSB-ResNet50 converges within 22 and 20 epochs across CPSCDB and MADB, which is consistent with theoretical analysis. ■

This proof provides the theoretical support for rapid convergence also validated by experimental results (99.4% ACC on ECG classification). For nonconvex optimization, our model converges to local minima or saddle points, yet achieves high practical performance.

TABLE IX

COMPARISON RESULTS OF PSB-RESNET50 WITH SEVERAL EXISTING STRATEGIES FOR LBBB AND RBBB DETECTION

Researchers	Year	Approach	Data set	Performance(%)
Matamoros et al.[29]	2020	BRNN+PCC	MADB	ACC=98.8
Bouny et al.[30]	2020	MA-CNN+Wavelet features	MADB	ACC=99.57
Tuncer et al.[25]	2020	Wavelet features+HHM	MADB	ACC=99.7 SEN=99.7
Yao et al.[31]	2020	Attention mechanism+CNN	CPSCDB	SEN=82.6 SPF=80.1
Jha et al.[26]	2020	Q-wavelet features+SVM	MADB	ACC=99.27 SEN=96.22
Singh et al.[33]	2022	ResNet	CPSCDB	ACC=97.9 SEN=98.6
Chandrasekar et al.[27]	2022	ECG features+Quadratic SVM	MADB	ACC=97.63
Zhang et al.[34]	2022	CNN+U-Net	CPSCDB	ACC=94.8 SEN=81.6
Zahid et al.[35]	2023	Self-operational network	MADB	ACC=99.2 SEN=99.1
Nahak et al.[28]	2023	ECG features+SVM	MADB	ACC=98.9
Sarankumar et al.[6]	2024	GRU+Auto-Encoder	CPSCDB	ACC=98.8 SEN=99.2
Din et al.[36]	2024	CNN+LSTM+Transformer	MADB	ACC=99.56 SEN=98.87
Cheng et al.[37]	2024	LFG-Net	CPSCDB	ACC=84.2
Yang et al.[3]	2025	Unet	CPSCDB	SEN=95.0
Xie et al.[45]	2025	CNN	MADB	ACC=93.45 SEN=90.84
This research	2026	PerC-SimAM-BLSTM	CPSCDB MADB	CPDB ACC=99.4 SEN=99.0 SPF=99.4
				MADB ACC=99.1 SEN=98.7 SPF=99.3

B. Complexity Analysis

To demonstrate the inference efficiency of the proposed PerC-SimAM-BLSTM, we reproduced several state-of-the-art classical strategies and implemented the same data processing and experimental configuration across CPSCDB. Also, the number of hyperparameters and floating-point operations (FLOPs) of network frameworks is computed to evaluate their time and space complexity as illustrated in Table VIII. Due to dense connection structure in CNN and self-attention mechanism in Transformer, the time and space complexity of PerC-SimAM-BLSTM are reduced by 43.5% and 36.8% compared with the CNN-Transformer [36] architecture. The experimental results confirm that the proposed PerC-SimAM-BLSTM has high inference efficiency and better potential for mobile devices. Similar conclusions are obtained on MADB.

C. Performance Comparison

Table IX elaborates on the main comparison results of PSB-ResNet50 with several existing strategies for LBBB and RBBB detection.

According to Table IX, recent research about RBBB and LBBB detection can be largely grouped into both conventional ML algorithm [24], [25], [26], [27], [28] and DCNN algorithm [29], [30], [31], [32], [33]. In terms of ML algorithm, applying the conventional ML approaches to ECG signal recognition is challenging due to some difficulties in inferior generalization ability, high-frequency occurrence of overfitting problem, and the absence of standardized guidelines for feature extraction. The issues are increasingly impeding their utilization in the actual implementation of many application scenarios.

Another research direction is to use DCNN models for ECG detection; some of them work depend on CNN blocks [33],

[45]; some of them resort to Transformer [36]; some of them seek help from self-operational network [35], Auto-Encoder [6], Unet [3], and LFG-Net [37]. Our model performances are higher than these algorithms except for the research [36] with an accuracy of 99.56%, but this model has a higher complexity as described in Section V-B. In summary, PerC-SimAM-BLSTM strikes a better balance between performance and complexity.

D. Limitations

Although the proposed PerC-SimAM-BLSTM demonstrates superior performance in detecting LBBB and RBBB, two principal limitations warrant attention in future work. First, the current validation is confined to LBBB, RBBB, and NSR across CPSCDB and MADB. To fully evaluate the generalizability, future studies need to extend the evaluation to other critical arrhythmias such as atrial fibrillation (AF) and ventricular tachycardia. AF detection, for instance, presents distinct challenges due to irregular R-R intervals and absence of P-waves, necessitating adaptations in temporal feature extraction. Additionally, real-world ECG data often contain motion artifacts, electrode noise, and signal drift, which are underrepresented in curated datasets. Validating the model on ambulatory ECG recordings and noisy clinical datasets would elucidate its robustness in practical scenarios. Techniques such as data augmentation with synthetic noise and transfer learning from curated to real-world domains could bridge this gap.

Second, despite the high performance of the model, the “uninterpretable” nature limits clinical interpretability. Future iterations should integrate attention visualization strategies (gradient-weighted class activation mapping) to highlight the regions of ECG waveforms that drive predictions. For instance, visualizing the PerC kernels and SimAM-BLSTM’s 3-D energy-based attention weights could reveal how global–local features interact to discriminate LBBB/RBBB. This aligns with clinical needs for explainable AI in diagnostics, enabling cardiologists to validate model decisions against electrophysiological principles. Implementing tools like attention overlays on raw ECG traces would further enhance trust and adoption.

VI. CONCLUSION

To overcome the inherent limitations of traditional convolutional receptive fields, we build PerC-SimAM-BLSTM, a novel MetaFormer-like model with a global receptive field to generate location-sensitive representations. This model not only inherits the global feature extraction ability of Transformer but also stretches the global–local feature acquisition ability of PerC. The effectiveness of PerC-SimAM-BLSTM is evaluated across RBBB and LBBB detection task. Experimental results achieve stable improvements over existing strategies and also strike a better balance between performance and complexity. The mathematical convergence proof further ensures stable convergence of model training to the theoretical optimal solution. Given its high performance, we explore the possibility of more application scenarios for the model, especially in lightweight devices.

REFERENCES

- [1] J. Wang, “An improved density-adaptive model based on motif matrix for myocardial infarction detection,” *Appl. Soft Comput.*, vol. 190, Mar. 2026, Art. no. 114551.
- [2] G. Ren, H. Liu, Y. Zhang, and Y. Li, “Attack detection based on encoding–decoding approach for cyber–physical systems,” *IEEE Trans. Syst., Man, Cybern., Syst.*, vol. 53, no. 10, pp. 6293–6302, Oct. 2023.
- [3] H. Yang, Y. Zhao, and X. Yang, “PearsonUnet: ECG classification model integrating Pearson related beats,” *IEEE Trans. Instrum. Meas.*, vol. 74, pp. 1–11, 2025.
- [4] Y. Shen et al., “AutoNet-generated deep layer-wise convex networks for ECG classification,” *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 46, no. 10, pp. 6542–6558, Oct. 2024.
- [5] J. Wang, “A modified state refinement GRU for atrial fibrillation and premature ventricular contraction signals detection,” *IEEE Trans. Syst., Man, Cybern., Syst.*, vol. 54, no. 12, pp. 7571–7579, Dec. 2024.
- [6] P. Singh and Y.-P. Huang, “AKDC: Ambiguous kernel distance clustering algorithm for COVID-19 CT scans analysis,” *IEEE Trans. Syst., Man, Cybern., Syst.*, vol. 54, no. 10, pp. 6218–6229, Oct. 2024.
- [7] K.-C. Huang, W. Lyu, M.-H. Yang, and Y.-H. Tsai, “PTT: Point-trajectory transformer for efficient temporal 3D object detection,” in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2024, pp. 14938–14947.
- [8] A. Anwar, Y. Khalifa, J. L. Coyle, and E. Sejdic, “Transformers in biosignal analysis: A review,” *Inf. Fusion*, vol. 114, Feb. 2025, Art. no. 102697.
- [9] J. Yu, J. Gao, N. Wang, P. Feng, B. Zhou, and Z. Wang, “Spa-tem MI: A spatial–temporal network for detecting and locating myocardial infarction,” *IEEE Trans. Instrum. Meas.*, vol. 72, pp. 1–12, 2023.
- [10] J. Kafková, B. Babusiak, R. Pirník, P. Kuchár, J. Kekelák, and F. D’Ippolito, “Seat to beat: Novel capacitive ECG integration for in-car cardiovascular measurement,” *Measurement*, vol. 240, Jan. 2025, Art. no. 115528.
- [11] A. Petroni, F. Cuomo, G. Scarano, P. Francia, M. Pediconi, and S. Colonnese, “MUSE: Multi-lead sub-beat ECG for remote AI based atrial fibrillation detection,” *J. Netw. Comput. Appl.*, vol. 212, Mar. 2023, Art. no. 103544.
- [12] J. Liu, C. Zhang, Y. Zhu, T. Ristaniemi, T. Parviainen, and F. Cong, “Automated detection and localization system of myocardial infarction in single-beat ECG using dual-Q TQWT and wavelet packet tensor decomposition,” *Comput. Methods Programs Biomed.*, vol. 184, Feb. 2020, Art. no. 105120.
- [13] K. He, X. Zhang, S. Ren, and J. Sun, “Deep residual learning for image recognition,” in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2016, pp. 770–778.
- [14] G. S. Bao et al., “Fast-detectGPT: Efficient zero-shot detection of machine-generated text via conditional probability curvature,” in *Proc. IEEE Intl. Conf. Learn. Represent.*, Vienna, Austria, Jun. 2024, pp. 1–14.
- [15] J. Wang, “BGcsSENet: Bidirectional GRU with spatial and channel squeeze-excitation network for bundle branch block detection,” *IEEE Trans. Hum.-Mach. Syst.*, vol. 53, no. 2, pp. 449–457, Apr. 2023.
- [16] H. Chen, X. Chu, Y. Ren, X. Zhao, and K. Huang, “PeLK: Parameter-efficient large kernel ConvNets with peripheral convolution,” in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2024, pp. 5557–5567.
- [17] A. Dosovitskiy et al., “An image is worth 16×16 words: Transformers for image recognition at scale,” in *Proc. IEEE Int. Conf. Learn. Represent.*, Vienna, Austria, Jun. 2021, pp. 1–12.
- [18] Z. Liu et al., “Swin transformer: Hierarchical vision transformer using shifted windows,” in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2021, pp. 9992–10002.
- [19] I. Tolstikhin et al., “MLP-Mixer: An all-MLP architecture for vision,” in *Proc. IEEE Comput. Vis. Pattern Recognit.*, Jun. 2021, pp. 1–14.
- [20] W. Yu et al., “MetaFormer is actually what you need for vision,” in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Kuala Lumpur, Malaysia, Jun. 2022, pp. 10809–10819.
- [21] H. K. Zhang, W. Z. Hu, and X. Y. Wang, “ParC-Net: Position aware circular convolution with merits from ConvNets and transformer,” in *Proc. IEEE Comput. Vis. Pattern Recognit.*, Kuala Lumpur, Malaysia, Jun. 2022.
- [22] L. X. Yang et al., “SimAM: A simple, parameter-free attention module for convolutional neural networks,” in *Proc. IEEE Int. Conf. Mach. Learn.*, Jun. 2021.
- [23] S. Sahoo, B. Kanungo, S. Behera, and S. Sabut, “Multiresolution wavelet transform based feature extraction and ECG classification to detect cardiac abnormalities,” *Measurement*, vol. 108, pp. 55–66, Oct. 2017.

- [24] H. Shi, H. Wang, F. Zhang, Y. Huang, L. Zhao, and C. Liu, "Inter-patient heartbeat classification based on region feature extraction and ensemble classifier," *Biomed. Signal Process. Control*, vol. 51, pp. 97–105, May 2019.
- [25] T. Tuncer, S. Dogan, P. Plawiak, and U. Rajendra Acharya, "Automated arrhythmia detection using novel hexadecimal local pattern and multilevel wavelet transform with ECG signals," *Knowl.-Based Syst.*, vol. 186, Dec. 2019, Art. no. 104923.
- [26] C. K. Jha and M. H. Kolekar, "Cardiac arrhythmia classification using tunable Q-wavelet transform based features and support vector machine classifier," *Biomed. Signal Process. Control*, vol. 59, May 2020, Art. no. 101875.
- [27] A. Chandrasekar, D. D. Shekar, A. C. Hiremath, and K. Chemmangat, "Detection of arrhythmia from electrocardiogram signals using a novel Gaussian assisted signal smoothing and pattern recognition," *Biomed. Signal Process. Control*, vol. 73, Mar. 2022, Art. no. 103469.
- [28] S. Nahak, A. Pathak, and G. Saha, "Fragment-level classification of ECG arrhythmia using wavelet scattering transform," *Expert Syst. Appl.*, vol. 224, Aug. 2023, Art. no. 120019.
- [29] A. Hernandez-Matamoros, H. Fujita, and H. Perez-Meana, "A novel approach to create synthetic biomedical signals using BiRNN," *Inf. Sci.*, vol. 541, pp. 218–241, Dec. 2020.
- [30] L. E. Bouny, M. Khalil, and A. Adib, "An end-to-end multi-level wavelet convolutional neural networks for heart diseases diagnosis," *Neurocomputing*, vol. 417, pp. 187–201, Dec. 2020.
- [31] Q. Yao, R. Wang, X. Fan, J. Liu, and Y. Li, "Multi-class arrhythmia detection from 12-lead varied-length ECG using attention-based time-incremental convolutional neural network," *Inf. Fusion*, vol. 53, pp. 174–182, Jan. 2020.
- [32] O. Faust and U. R. Acharya, "Automated classification of five arrhythmias and normal sinus rhythm based on RR interval signals," *Expert Syst. Appl.*, vol. 181, Nov. 2021, Art. no. 115031.
- [33] P. Singh and A. Sharma, "Interpretation and classification of arrhythmia using deep convolutional network," *IEEE Trans. Instrum. Meas.*, vol. 71, pp. 1–12, 2022.
- [34] X. Zhang, H. Wu, T. Chen, and G. Wang, "Automatic diagnosis of arrhythmia with electrocardiogram using multiple instance learning: From rhythm annotation to heartbeat prediction," *Artif. Intell. Med.*, vol. 132, Oct. 2022, Art. no. 102379.
- [35] M. U. Zahid, S. Kiranyaz, and M. Gabbouj, "Global ECG classification by self-operational neural networks with feature injection," *IEEE Trans. Biomed. Eng.*, vol. 70, no. 1, pp. 205–215, Jan. 2023.
- [36] S. Din, M. Qaraqe, O. Mourad, K. Qaraqe, and E. Serpedin, "ECG-based cardiac arrhythmias detection through ensemble learning and fusion of deep spatial-temporal and long-range dependency features," *Artif. Intell. Med.*, vol. 150, Apr. 2024, Art. no. 102818.
- [37] Y. Cheng, D. Li, D. Wang, Y. Chen, and L. Wang, "Multi-label arrhythmia classification using 12-lead ECG based on lead feature guide network," *Eng. Appl. Artif. Intell.*, vol. 129, Mar. 2024, Art. no. 107599.
- [38] F. Liu et al., "An open access database for evaluating the algorithms of electrocardiogram rhythm and morphology abnormality detection," *J. Med. Imag. Health Informat.*, vol. 8, no. 7, pp. 1368–1373, Sep. 2018.
- [39] G. B. Moody and R. G. Mark, "The impact of the MIT/BIH arrhythmia database," *IEEE Eng. Med. Biol. Mag.*, vol. 20, no. 3, pp. 45–50, May 2001.
- [40] M. Amrouche and D. M. Stipanovic, "A formal characterization of activation functions in deep neural networks," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 35, no. 2, pp. 2153–2166, Feb. 2024.
- [41] B. De Weerd, Y. C. Eldar, and N. Deligiannis, "Deep unfolding transformers for sparse recovery of video," *IEEE Trans. Signal Process.*, vol. 72, pp. 1782–1796, 2024.
- [42] X. Lin et al., "Context-gated convolution," in *Proc. Eur. Conf. Comput. Vis.*, 2020.
- [43] S. A. Hillyard, E. K. Vogel, and S. J. Luck, "Sensory gain control (amplification) as a mechanism of selective attention: Electrophysiological and neuroimaging evidence," *Phil. Trans. Roy. Soc. London. Ser. B, Biol. Sci.*, vol. 353, no. 1373, pp. 1257–1270, Aug. 1998.
- [44] J. Hu, L. Shen, S. Albanie, G. Sun, and E. Wu, "Squeeze-and-excitation networks," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 42, no. 8, pp. 2011–2023, Aug. 2020.
- [45] Y.-L. Xie and C.-W. Lin, "YOLO-ResTinyECG: ECG-based lightweight embedded AI arrhythmia small object detector with pruning methods," *Expert Syst. Appl.*, vol. 263, Mar. 2025, Art. no. 125691.

Jibin Wang received the degree from the School of Mathematics, Tianjin University, Tianjin, China.

He is currently with Anhui Key Laboratory of Digital Medicine and Intelligent Health, School of Medical Imaging, Bengbu Medical University, Bengbu, China. His research focuses on deep learning and ECG signal analysis.

Bo Shi received the B.S. degree in electronic science and technology from Xuzhou Normal University, Xuzhou, China, in 2004, and the M.S. degree in biomedical engineering from Nanjing University of Science and Technology, Nanjing, China, in 2006.

He is currently a Professor with Anhui Key Laboratory of Digital Medicine and Intelligent Health, School of Medical Imaging, Bengbu Medical University, Bengbu, China. His research interests include novel biomedical instrumentation, physiological measurements, ECG signal analysis, and medical AI.